

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Paper No. BSCU7A: Ancillary Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **six** of the following: [6 × 3 = 18]

- (a) What do you understand by streamline flow of a liquid?
- (b) What is the difference between stress and strain?
- (c) What should be done to reduce the surface tension of water?
- (d) Define the entropy of a system.
- (e) Give one example of the first law of thermodynamics.
- (f) What are the advantages and disadvantages of CO₂ gas in the environment?
- (g) How is the image of an extended source formed by a convex lens?
- (h) What do you understand by the dispersion of light?

Question 2.

- (a) What do you mean by Hooke's law? Describe Young's modulus of elasticity. [7]
- (b) Discuss Bernoulli's theorem and its applications. [6]

Question 3.

- (a) Define the surface tension of a liquid. Describe in detail the capillary action of a tube. [7]
- (b) What is Stokes' law? Discuss the terminal velocity of a sphere falling in a fluid. [6]

Question 4.

- (a) Discuss the second law of thermodynamics and its importance. [7]
- (b) Discuss the phenomenon of diffusion of gases. On what factors does it depend? [6]

Question 5.

- (a) Define Helmholtz and Gibbs free energies with some examples. [7]
- (b) Discuss Fermat's principle of least time. [6]

Question 6.

- (a) Discuss Young's double-slit experiment and the conditions for maximum and minimum intensity. [7]
- (b) What is the difference between Fraunhofer and Fresnel diffraction? [6]

Question 7.

- (a) Discuss the effect of radiation on biological systems. [7]
- (b) Discuss all parts of the electromagnetic spectrum. [6]